Folding Chair Teacher Guide

Using the Folding Chair workbook

Student edition: 50 A4 pages, printed double-sided on the long edge as 25 sheets. Use the PDF for controlled printing. The Word edition contains editable text, tables and working areas; changes can affect pagination.

The workbook follows all 58 named theory sections in the ten paired-week website modules. Its 30 multiple-choice questions are selected from the course's 120-question bank; the remaining online questions and 40 scaffolded written responses stay available on the site. Paper applications are formative learning activities, not a new formal assessment.

Teach the matching section, model one example, then ask students to attempt the paper task before discussing the guide. Use pages 47–50 throughout making. Students should date evidence, identify components and distinguish actual observations from proposed or hypothetical results.

Machine use, workshop demonstrations, product selection and practical inspection remain teacher-controlled. Supply the current SOP, actual product label and current exact-product SDS where required. No workbook entry certifies a chair as safe or authorises a load test.

Authority, sources and teacher decisions

The controlling construction document is Folding-Chair-Project-Plans.pdf: four original sheets dated 2 December 2020, reproduced unchanged in information on student pages 43–46. Written dimensions and notes control manufacture; do not scale the resized reproductions. Materials and finish fields are blank on those plans.

Teacher to confirm the actual stock, preparation allowances, adhesive/finish product and instructions, equipment access and local project placement. The current course identifies Folding Chair as a provisional Stage 5 project resource and its program as a local teacher review copy. Confirm formal assessment conditions, due dates and submission requirements separately.

Mortise-and-tenon content teaches joinery principles and practice; it does not substitute that joint for the plan's trenches, rebate or moving pivots. Distinguish the 6.5 mm holes from M6 × 50 hardware and preserve mirrored countersink instructions. A source model mentioning ten folding cycles is an illustrative response, not a prescribed test.

Source snapshot: stevencowell/folding-chair-guided-course, revision 7877feafbbe391eba4ff2e9af18494ac650fef6f, retrieved for this build on 12 September 2026. Course: https://stevencowell.github.io/folding-chair-guided-course/. Teacher resources: https://stevencowell.github.io/folding-chair-guided-course/teacher-resources.html. Plans: https://stevencowell.github.io/folding-chair-guided-course/Folding-Chair-Project-Plans.pdf.

Original explanatory text adapts the authorised course. Source questions retain wording and option order. Bespoke photographs are AI-generated teaching illustrations, not actual student work; conceptual schematics carry no project dimensions. Design comparisons used Mirror Edition 2.0, Footstool and Desk Tidy solely for workbook finish and structure.

Student page 3

Explain why both the open and folded positions belong in your design brief.

The open chair must be stable and structurally sound; the folded chair needs controlled clearance and movement. A design must satisfy both states, so a decorative or dimensional change must be checked for both.

Source: Module 1, section 1; Module 1, section 2

Student page 4

Identify one hazard at a moving chair pivot, describe the possible harm and propose a suitable control.

A finger can enter a closing gap and be pinched. Identify the movement path and pinch point, keep hands clear during teacher-directed checking and stop when movement or clearance is uncertain. Accept another specific hazard–harm–control chain grounded in the source.

Source: Module 1, section 3

Student page 5

One student measures a hole from a squared datum end; another starts from a hand-shaped curve. Their rule readings match. Explain why their hole centres may still differ.

Equal rule readings do not establish equal physical positions when the starting references differ. The curve is an uncertain reference; recheck both centres from the same verified datum using the written plan dimensions, then directly compare the pair before drilling.

Source: Module 1, section 4; Module 1, section 5; Module 1, section 6

Student page 6

1. The plan is displayed smaller on your screen than it was when it was drawn. Which information should control the chair dimensions?

B. The written dimensions and notes on the drawing

Written dimensions remain authoritative even if the drawing has been enlarged or reduced.

Source: Module 1 knowledge checks

2. Why are washers placed between moving timber components at a pivot?

B. To separate the faces, spread pressure and reduce friction and wear

Washers separate the surfaces, distribute pressure and help the joint rotate without excessive wear.

Source: Module 1 knowledge checks

3. The drawing says to countersink left- and right-hand components as pairs. What does this mean?

C. Arrange the parts as mirror images and countersink the intended matching outside faces

Mirrored parts are laid out together so the countersinks occur on the correct opposite faces.

Source: Module 1 knowledge checks

Explain a reliable marking routine for a mirrored pair, from reference surfaces to the final pre-cut check.

Establish and mark face side, face edge and squared end; measure related features from one datum; rule flat and eye above graduation; use fine marks and identify waste; lay parts in assembly orientation and label left/right; cross-check sizes, pivot centres and countersink faces against plans before removing material.

Source: Module 1, section 1; Module 1, section 2; Module 1, section 3; Module 1, section 4; Module 1, section 5; Module 1, section 6

Student page 7

A knot is near a proposed pivot centre. What information should you check before accepting that board?

Check the actual component layout, defect extent, fibre direction, source-plan pivot position and remaining sound material; seek the teacher's allocation decision. Do not shift an authorised pivot just to avoid a knot. Accept suitable alternative stock or an approved reallocation.

Source: Module 2, section 1; Module 2, section 2

Student page 8

Why can preparing final length before a square datum end produce a misleading result?

The length can differ along opposite edges when the end is not square. Establishing the squared datum end first makes final length repeatable and supports consistent paired components.

Source: Module 2, section 3

Student page 9

Why is adding glue a poor response to a joint that has not passed its dry fit?

Glue cannot correct a dimensional or alignment fault. It can hide the cause, make disassembly difficult and leave poor surface contact or split timber if the joint is forced. Diagnose and correct the dry fit first.

Source: Module 2, section 4; Module 2, section 5

Student page 10

1. What does DAR mean, and what should a careful maker still do before using it?

B. Dressed All Round; inspect it for movement, defects, size and squareness

DAR means Dressed All Round, but the board must still be checked for movement, defects, size and squareness.

Source: Module 2 knowledge checks

2. Why can a thicknesser fail to remove twist from an undressed board?

B. It makes the upper face parallel to the surface supported on the bed, so an unstable underside can reproduce the distortion

A thicknesser produces a face parallel to the supported underside; without a flat reference, twist or cup may remain.

Source: Module 2 knowledge checks

3. Why are bolts, washers and locknuts used at the chair pivots instead of gluing a mortise-and-tenon joint there?

A. The pivot must allow controlled rotation

The pivot requires controlled movement, so mechanical hardware is used rather than a fixed glued joint.

Source: Module 2 knowledge checks

Put FEWTEL in order, then explain how its first two references protect later pivot marking and assembly.

Face, Edge, Width, Thickness, End, Length. The flat face and square straight edge create consistent references for parallel sizing, a squared end and accurate feature locations. Check actual flatness and squareness; matching numeric sizes alone cannot remove twist or misalignment.

Source: Module 2, section 1; Module 2, section 2; Module 2, section 3; Module 2, section 4; Module 2, section 5

Student page 11

Explain why a curved cut should retain the finished line and why a relief cut must remain in waste.

Retaining the line leaves controlled material for refining saw marks without making the component undersize. A relief cut is only for releasing waste; crossing the line damages the finished component and cannot be repaired by sanding.

Source: Module 3, section 1; Module 3, section 2

Student page 12

Sequence puzzle: number these actions after unusual vibration: A seek teacher help; B stop feeding; C wait for the blade to stop; D switch off as directed. Explain why cutting must not continue.

B → D → C → A describes the controlled stop and report sequence in the source. Alerting the teacher may occur immediately as well; do not penalise a safe early call for help. Changed vibration may signal loss of control or equipment fault; continued cutting increases risk.

Source: Module 3, section 3

Student page 13

Choose one observation above. Explain the cause you would investigate and what would count as a successful recheck.

Accept a specific check linked to the symptom: compare profiles/lengths and references; identify a local high spot or alignment fault; restore a verified datum rather than guessing. A successful recheck supplies measured/template or fit evidence, not just appearance.

Source: Module 3, section 4; Module 3, section 5

Student page 14

1. What is the safest response if a marked curve is too tight for the fitted bandsaw blade?

B. Stop and use an approved blade, jig or alternative process

Changing to an approved method preserves safety and accuracy.

Source: Module 3 knowledge checks

2. Which condition requires the operator to stop feeding and switch off as directed?

B. The timber lifts and the blade begins to stall

Lifting work and a stalling blade indicate a serious loss of control; stop and seek assistance.

Source: Module 3 knowledge checks

3. Why should paired R32 or R16 profiles be compared directly after shaping?

B. To confirm matching length, curve and transition before later drilling

Direct comparison reveals mismatched profiles while correction is still practical.

Source: Module 3 knowledge checks

A curve is tighter than the fitted bandsaw blade can follow. Explain the safe response and how you would protect the finished profile.

Stop feeding and stop as directed; wait for full stop and consult the teacher about blade/process or relief-cut alternatives. Never twist or force the blade. Reconfirm the approved profile/waste and leave the finished line for controlled refinement. No student changes to blade setup without authorisation.

Source: Module 3, section 1; Module 3, section 2; Module 3, section 3; Module 3, section 4; Module 3, section 5

Student page 15

A hole enters on the marked centre but exits off-centre. Explain what was and was not proved by the entry mark.

The entry centre may be correct, but perpendicularity and stable setup are not proved. Check work support, table/fence/bit alignment and movement with the teacher; inspect the exit and compare the axis before accepting or correcting the part.

Source: Module 4, section 1; Module 4, section 2

Student page 16

Explain why “the locknut is fitted” does not prove the pivot is correctly adjusted.

A locknut resists loosening but may still be overtightened or left too loose. Check washer positions, head seating, movement through the required range and excessive play under teacher supervision.

Source: Module 4, section 3; Module 4, section 4

Student page 17

Write one hold point for your drilling-to-assembly sequence. State the evidence required and the action if it fails.

For example, before countersinking/assembly, compare paired pivot centres, perpendicularity and intended faces against the plan. If alignment fails, stop the sequence, record the fault and seek teacher-approved diagnosis/correction; retest before proceeding.

Source: Module 4, section 5; Module 4, section 6

Student page 18

1. Why does the chair drawing specify a 6.5 mm hole for an M6 pivot bolt?

A. To create a small clearance for assembly and rotation

The 6.5 mm hole provides a small clearance around the 6 mm bolt for assembly and movement.

Source: Module 4 knowledge checks

2. Why is a washer placed between the moving timber members?

B. To separate bearing faces, spread pressure and reduce friction and wear

It separates the timber faces, distributes pressure and reduces direct friction.

Source: Module 4 knowledge checks

3. Which item belongs in a useful WMS for drilling and assembling the chair pivots?

B. Sequence, hazards, controls, quality checks and response if a fault appears

A useful WMS coordinates the task, controls and hold points.

Source: Module 4 knowledge checks

The chair binds during dry assembly. Describe a reasoned diagnostic sequence before any timber is removed.

Locate the point of resistance and contacting surfaces; inspect washer stack, bolt-head seating, teacher-approved tension, component twist and pivot alignment; separate/check as directed. Identify the actual cause, choose the least invasive approved correction, then retest the full range and record the outcome.

Source: Module 4, section 1; Module 4, section 2; Module 4, section 3; Module 4, section 4; Module 4, section 5; Module 4, section 6

Student page 19

Choose one permitted design detail. Identify two competing criteria and the evidence you would need before recommending it.

Accept a realistic detail within the teacher-approved scope, two distinct relevant criteria and evidence for both. For example edge treatment: handling comfort versus remaining section/clearance, checked on approved sketch/sample and movement review. Do not accept unauthorised pivot relocation.

Source: Module 5, section 1; Module 5, section 2

Student page 20

A design has the highest matrix score but collides when folded. Explain your decision and the records that need updating after an approved revision.

Reject or revise the collision-producing option: the safety/function requirement is not traded away by a total score. Recheck the cause and alternatives; after teacher approval update the sketch/CAD views, dimensions/notes, cutting list, production method and revision identity consistently.

Source: Module 5, section 3; Module 5, section 6

Student page 21

On the concept diagram, explain why the front hole is circular but the corresponding top and side features use hidden lines.

The front looks along the through-hole axis and sees its circular opening. The top and side look across the solid block, so the bore edges are concealed and shown with hidden lines. Corresponding widths/heights align between views.

Source: Module 5, section 4; Module 5, section 5

Student page 22

1. In third-angle projection, where is the top view placed?

B. Above the front view

In third-angle projection the top view is above the front view.

Source: Module 5 knowledge checks

2. What does “6.5 mm Ø” mean?

B. A circular hole with a diameter of 6.5 mm

The notation specifies a 6.5 mm diameter circular hole.

Source: Module 5 knowledge checks

3. Why must a CAD drawing still be checked against the project plan?

A. CAD can draw a wrong dimension very accurately

A perfectly drawn but incorrectly entered dimension remains a manufacturing error.

Source: Module 5 knowledge checks

Explain how you would check a revised CAD drawing before it guides manufacture. Include units, views, references and version control.

Confirm millimetres, approved plan/design scope, datum origins and critical values; compare front/top/right-side alignment in third-angle, hidden/centre/visible line conventions and notes; check mirror faces and movement clearance; resolve contradictions and identify current revision; update dependent cutting list/production records after teacher approval.

Source: Module 5, section 1; Module 5, section 2; Module 5, section 3; Module 5, section 4; Module 5, section 5; Module 5, section 6

Student page 23

A student finds that the last rail blocks access to a clamp. Explain one change to the rehearsal and why extra adhesive would not solve this problem.

Rehearse a different approved subassembly or order so the joint and clamp remain accessible; confirm dry fit before applying adhesive. Extra adhesive cannot restore clamp access or correct a poorly fitted joint. Accept another source-consistent change that preserves the approved construction and access.

Source: Module 6, section 1; Module 6, section 2

Student page 24

Write a brief correction note for each error: "More clamp pressure will fix the unequal diagonals" and "A price per metre can be multiplied directly by a length in millimetres."

For the frame, locate the racking and adjust clamp position or pressure in a controlled way, then remeasure; do not add force everywhere. For costing, convert millimetres to metres before multiplying by a price per metre. The note must connect each correction to reliable evidence, not merely mark it wrong.

Source: Module 6, section 3; Module 6, section 4

Student page 25

A subassembly needs longer before the next operation, and finishing is planned next. Write one useful parallel task and one hold point that should remain in the revised plan.

A valid parallel task is updating authentic folio captions, quantities, time records or the evaluation plan. Retain the inspection after the required clamp/cure stage and the function/surface check before finishing. Accept another independent task if it does not disturb the assembly or bypass required preparation. Do not accept shortened product waiting times.

Source: Module 6, section 5; Module 6, section 6

Student page 26

Why should a complete dry fit occur before adhesive is opened?

A. To confirm fit, sequence, clamp access and measurements while correction is still easy

Dry fitting verifies the method and exposes faults before the glue-up becomes time-critical.

Source: M06 mc01

A timber component is 450 mm long and costs $8.00 per metre. What is the material cost for one component before waste allowance?

B. $3.60

$3.60 is 0.45 m multiplied by $8.00 per metre.

Source: M06 mc07

What is the critical path?

A. The chain of dependent tasks that controls the earliest completion date

The critical path controls overall completion through linked dependent tasks.

Source: M06 mc11

Hypothetical production problem: the dry-fit frame is racked, the adhesive is ready, and the plan has fallen behind. Explain a revised sequence that protects fit and adhesive timing. Include a hold point, a useful parallel task and one costing record that may need updating.

Expected evidence: stop before adhesive application and diagnose the racked dry fit; adjust the approved assembly/clamp arrangement and recheck consistent diagonals, squareness and joint closure; prepare the work area and work within the actual product open time; retain required clamp/cure periods and inspect at the hold point before later stages. A parallel task can be folio, costing or time-record updates while the assembly is undisturbed. Update actual quantities, consumables, waste allowance or approved variations if changed, and explain time variance. Accept alternative source-consistent sequencing; no invented adhesive time, approval or school price.

Source: Module 6, section 1; Module 6, section 2; Module 6, section 3; Module 6, section 4; Module 6, section 5; Module 6, section 6

Student page 27

A rail feels smooth but still has one deep cross-grain scratch and a rounded edge. Explain why moving immediately to finer paper is a poor next step.

Fine abrasive may leave the deeper scratch and will not restore the rounded geometry. Diagnose both defects, seek the approved correction where shape has changed, and use a suitable controlled abrasive stage to remove the scratch without unnecessary material loss. Reinspect before progressing. No fixed new grit or dimension is required.

Source: Module 7, section 1; Module 7, section 2

Student page 28

Explain why both the matching-scrap test and the cure requirements matter to a folding chair. Include one appearance issue and one movement issue.

Matching scrap shows the complete system's colour, grain/end-grain response or sheen before committing to the chair. Cure requirements matter because touch-dry coating may mark or stick under pressure; film build at moving faces can also reduce clearance. The answer should connect appearance and movement to evidence, without inventing product times.

Source: Module 7, section 3; Module 7, section 4

Student page 29

Choose one fault above. Write one quality-control observation and one quality-assurance change that could reduce its recurrence.

Examples: for dust nibs, inspect and record raised particles after the appropriate drying stage; improve cleaning, applicator preparation or dust control before later coats. For a pale patch, record its location near a joint and improve glue-residue inspection before coating. For a sag, record thickness/location and improve controlled thin application. Accept justified alternatives; causes remain hypotheses until checked.

Source: Module 7, section 5; Module 7, section 6

Student page 30

Why should dried glue be removed before normal finish sanding?

A. It can block finish absorption and appear as a pale patch

Glue contamination can prevent even wetting or penetration and remain visible.

Source: M07 mc01

What is the main difference between drying and curing?

A. Drying is an early handling stage; curing is the longer development of hardness and resistance

The distinction explains why apparently dry coating may still mark or stick.

Source: M07 mc06

What is quality assurance?

A. Planned systems that prevent defects and ensure requirements are built into the process

Quality assurance is process-focused and preventive.

Source: M07 mc10

Hypothetical first coat: dust nibs cover one face, a sag appears on a rail and a pale patch sits beside a joint. Explain likely causes and a controlled response. Include one preventive change for the next coating session and explain why a further coat should wait.

Expected evidence: dust nibs may arise from airborne/surface/applicator contamination; a sag suggests heavy or uneven application; a local pale patch may be residual glue or uneven preparation. Record observations, identify the exact product and its current state, and seek the approved correction after the required drying/cure stage. Do not immediately add a coat that may trap contamination, deepen the sag or hide the diagnosis. Preventive changes can improve dust control, applicator cleaning, glue-residue checks, thin application or staged inspection. Accept plausible alternatives with cautious causal language and product-specific teacher/manufacturer direction.

Source: Module 7, section 1; Module 7, section 2; Module 7, section 3; Module 7, section 4; Module 7, section 5; Module 7, section 6

Student page 31

For planning only: name the SDS section you would use to check storage and the section for glove selection. Explain why the exact product name still has to be checked first.

Start with Section 7 for handling and storage, cross-checking Section 10 for incompatibilities where relevant. Start with Section 8 for glove/exposure-control requirements. Check identification in Section 1 because products or suppliers can differ in composition, hazards and required controls. Use the current teacher-confirmed exact-product SDS; this is a navigation exercise.

Source: Module 8, section 1; Module 8, section 2

Student page 32

Explain two flaws in this plan: “I will use gloves as the only vapour control, then add polish to make a stiff pivot move.”

Gloves do not control inhaled vapour and PPE alone bypasses stronger controls such as suitable product selection, isolation/ventilation and the approved procedure. A stiff pivot requires mechanical diagnosis; polish could add incompatible residue without addressing the cause. Exact PPE and maintenance product are teacher/product dependent.

Source: Module 8, section 3; Module 8, section 4

Student page 33

Choose either material yield or repairability. Explain one realistic improvement and one condition that must remain satisfied for it to be a sustainable choice.

Yield example: plan component placement and use suitable short defect-free sections to reduce offcuts while preserving approved dimensions, grain suitability and sound material. Repairability example: retain sound members and replace specified worn hardware, provided the surrounding timber and original geometry remain sound and the repair is assessed. Accept another evidence-based trade-off; do not accept unsupported thinning or a claim that all wood is sustainable.

Source: Module 8, section 5; Module 8, section 6

Student page 34

Which SDS section is the first place to look for eye-exposure first aid?

A. Section 4

Section 4 contains first-aid measures for exposure routes.

Source: M08 mc03

Why is PPE placed near the bottom of the hierarchy of controls?

A. It depends on correct selection, fit and use and does not remove the hazard

PPE reduces exposure but does not eliminate the source and relies on consistent use.

Source: M08 mc06

How does standard replaceable pivot hardware support sustainability?

A. Worn hardware can be inspected and replaced without discarding the whole chair

Replaceable hardware supports maintenance and longer use.

Source: M08 mc11

Hypothetical maintenance plan: a chair has a dull surface and a stiff pivot. Explain how you would decide whether cleaning, compatible polishing or mechanical assessment is needed. Include the current product information and controls required, then explain how a sound repair could reduce waste.

Expected evidence: diagnose dullness and stiffness separately; establish the exact finish and appropriate cure/maintenance information rather than applying an unknown polish; consult the current exact-product label/SDS supplied by the teacher and the school procedure; choose stronger suitable exposure controls before relying on selected PPE. Inspect the mechanism and report abnormal resistance rather than forcing it or lubricating indiscriminately. A suitable assessed repair may retain sound timber and replace specified worn hardware, reducing whole-product replacement, but damaged holes/timber or unsafe geometry may prevent repair. Accept justified alternatives; do not fabricate product details or disposal instructions.

Source: Module 8, section 1; Module 8, section 2; Module 8, section 3; Module 8, section 4; Module 8, section 5; Module 8, section 6

Student page 35

Replace "This is my chair" with a two-sentence caption for a real or clearly labelled hypothetical detail. Include the stage, evidence and next action or verified outcome.

Criteria: identifies a relevant stage and feature; states what the image/check can actually prove; includes a genuine outcome or an explicitly pending next action. A hypothetical response must be labelled. Accept details about paired shape, dry-fit joint closure, hardware, surface preparation or another authentic project feature. Do not credit invented teacher approval or borrowed results.

Source: Module 9, section 1; Module 9, section 2

Student page 36

Close the reading. Write from memory the difference between description and evaluation, then add one chair-specific example. Reopen it and correct one omission if needed.

Description states characteristics or what occurred. Evaluation judges against criteria using relevant evidence and limitations, usually explaining causes and improvement. A valid example contrasts "it folds" with an evidence-based judgement from a real check, without inventing results. Student corrections count as useful retrieval evidence; the example may explicitly remain a sentence frame if no practical evidence exists.

Source: Module 9, section 3; Module 9, section 4

Student page 37

For "Explain how accurate folding action depends on manufacture", draft a three-link answer plan. Name a principle, its chair application and a consequence.

Example: a consistent datum controls measurement; paired frame components use the reference for mark-out and pivot locations; matching geometry supports aligned holes and clear movement at dry assembly. Other valid chains may connect paired shaping, drilling checks, hardware arrangement and movement. Require cause and effect rather than three isolated process names; do not invent tolerances or teacher approval.

Source: Module 9, section 5; Module 9, section 6

Student page 38

What should a strong photograph caption include?

A. The stage, process or decision, evidence and outcome

These elements make the evidence traceable and meaningful.

Source: M09 mc02

What is retrieval practice?

A. Trying to recall information before checking the answer

Retrieval strengthens access to knowledge and reveals gaps.

Source: M09 mc06

A long-response question asks how accurate folding action is achieved. Which answer plan is strongest?

A. Connect datums, prepared reference surfaces, paired shaping, pivot drilling and dry assembly with cause and effect

This plan connects the relevant stages and explains how they support the outcome.

Source: M09 mc11

Choose a real project stage, or label a hypothetical one. Write a short evaluation paragraph that states a criterion, identifies evidence, makes a judgement and proposes a specific improvement. Finish with one retrieval question you could use later to check the principle behind that improvement.

Criteria: the chosen stage is relevant and any hypothetical evidence is identified; criterion is from the brief/approved requirements or clearly framed as a learning example; evidence is specific and authentic, not a claimed unperformed test; judgement explains how well the criterion is met and any limitation; improvement addresses a plausible cause and can be verified. The retrieval question should require recall of the principle, for example why a consistent datum supports paired pivot accuracy, not just "Did I do well?". Accept a strength-based evaluation where no fault is evidenced. No invented marks, results or sign-off.

Source: Module 9, section 1; Module 9, section 2; Module 9, section 3; Module 9, section 4; Module 9, section 5; Module 9, section 6

Student page 39

A pivot binds only near the folded position. Explain why sanding the frame immediately is a weak response, and identify two observations to record first.

Immediate sanding removes sound material before the cause is known and may create excess clearance or side play. Record the exact position/contact where binding begins and the condition or seating of the specified hardware/washer stack. Other useful observations include coating contact, local gap change, alignment or twist. The next correction needs teacher direction and a repeated full check; no unapproved loads.

Source: Module 10, section 1; Module 10, section 2

Student page 40

Write a two-sentence handover note: one sentence reports a hypothetical equipment fault without diagnosing beyond the evidence; the other identifies a portfolio check before submission.

A valid hypothetical report identifies equipment, an observable condition and the operation/time context, with a stop/report response rather than unauthorised repair. Example: "During the last approved operation, I noticed unusual vibration at [machine] and stopped to report it." The second sentence names a real check such as current revisions, captions, quantities/totals, authentic evidence or evaluation against criteria. Hypothetical fault must remain labelled.

Source: Module 10, section 3; Module 10, section 4

Student page 41

Choose one practice to keep or change. Explain how you will verify it next time, then add one care instruction that protects the finished chair's movement or timber.

Criteria: a specific practice linked to actual evidence or clearly stated planning; a feasible verification such as a recorded paired-part check, dry-fit inspection or caption; and a valid care instruction, for example inspect hardware and timber, store dry, do not force a stiff mechanism, or remove an unstable chair from use for assessment. Do not require invented defects, load ratings, dates or maintenance products.

Source: Module 10, section 5; Module 10, section 6

Student page 42

What does "least invasive correction" mean?

A. Restore the requirement while changing or removing the least sound material

A least invasive action corrects the verified cause with minimal collateral change.

Source: M10 mc04

What is an as-built record?

A. Documentation of the product as it was actually completed, including approved differences

An as-built record documents the final product and authorised changes.

Source: M10 mc07

What is the correct response if the chair becomes unstable during later use?

A. Remove it from use until the cause is assessed and repaired appropriately

Removing the chair from use prevents exposure while diagnosis occurs.

Source: M10 mc11

Prepare a concise final handover note for your chair. Include a completed or pending inspection, an unresolved issue or verified strength, the matching portfolio evidence, one workshop handover action and one specific care instruction. Keep unknown details visibly pending rather than inventing a successful check.

Criteria: clearly distinguishes completed evidence from pending checks; identifies a meaningful structural, hardware, movement, stability or finish inspection within teacher-approved procedures; names an honest limitation or verified strength; links to a matching caption, as-built record, costing or evaluation; includes a suitable workshop action such as tool return, fault report, clear access or correct waste/chemical storage; and provides compatible care advice such as dry storage, inspection, avoiding forced movement or removal from use if unstable. No invented load test, pass, teacher approval, formal submission date or maintenance product.

Source: Module 10, section 1; Module 10, section 2; Module 10, section 3; Module 10, section 4; Module 10, section 5; Module 10, section 6

Student page 47

Compare the concepts against two criteria. Which do you recommend, and why?

Look for two meaningful criteria, evidence-based comparison and a reasoned trade-off. Source-plan structure, pivots and dimensions remain controlling; any proposed change requires the teacher's approval.

Source: Module 1, section 4; Module 5, section 1; Project folio; Original project plan

Concept A

Assess clear labelled sketch, source constraints, the proposed variation and a useful explanation. Alternatives are valid; this is design evidence, not one prescribed drawing.

Source: Project folio; Module 1, section 4

Concept B

Assess clear labelled sketch, source constraints, the proposed variation and a useful explanation. Alternatives are valid; this is design evidence, not one prescribed drawing.

Source: Project folio; Module 1, section 4

Student page 48

Record the actual timber, one inspection finding and your component-allocation decision.

Look for the actual teacher-confirmed stock rather than a guessed species; an observed grain/defect/distortion condition; and a reason for using, reallocating or rejecting the piece. Link the decision to the proposed component and teacher direction.

Source: Project folio

Record the datum and pair/handedness check you completed before cutting.

Identify verified face side, face edge and squared end; source-plan dimensions, left/right labels and correct faces. Evidence may be a dated comparison, annotated sketch or photo reference; do not accept a claim of accuracy without the check.

Source: Project folio

Component/detail sketch

Assess alignment, correct feature/datum identification, line conventions and faithful written dimensions. The student must not estimate from the resized picture. Source-plan figures may differ by selected component.

Source: Original project plan; Project folio

Three cutting-list entries

Check names, quantities, finished dimensions and stock/allowance notes against the actual plan and teacher decision. Keep cross members subject to the source custom-fit instruction after trenches. Do not invent allowances or interpret compressed extracted number strings as dimensions.

Source: Original project plan; Project folio; Module 2, section 1

Student page 49

Caption: date and stage → what you did → evidence checked → correction or result.

Look for authentic evidence, exact stage/component, relevant process and quality check, a reasoned correction where applicable and an observed outcome. Identify assistance and sources when required. A useful caption interprets the image instead of repeating "this is my chair".

Source: Project folio

Name a hazard from this stage and the control you actually used.

Require a specific stage-relevant hazard, possible harm and actual control, consistent with teacher-authorised work. Do not accept invented demonstrations or PPE alone where stronger controls are necessary.

Source: Project folio

Process image/sketch

Assess relevance and authenticity, labels showing the feature being checked, and clear connection to the caption. An illustrative picture from this workbook is not evidence of the student's practical performance.

Source: Project folio; Module 9, section 1

Production checkpoint record

Each row needs an actual observation or evidence reference: joint closure/squareness, source hardware and movement, slat spacing/alignment/support, then preparation/product/stage of finish. Mark unresolved issues honestly and follow teacher-directed checking; do not infer a safe load from appearance.

Source: Project folio; Module 6, section 3; Module 4, section 4; Module 7, section 1

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Choose one success criterion. Give your evidence, a judgement, a limitation and one specific improvement.

Require a clearly named criterion, authentic observation or teacher-directed check, a supported judgement and limitation. The proposed improvement must address the identified cause. No invented load/cycle results or claim of teacher sign-off.

Source: Project folio

Which working habit will you carry into the next project? Explain what happened, why it mattered and how you will verify the improvement.

Accept a specific transferable habit such as datum checks, rehearsed assembly or early fault reporting; link experience to cause, a practical next action and a way to check its effect. Generic “work harder” is not sufficient.

Source: Project folio

Record the teacher-confirmed care advice and any unresolved handover issue.

Use actual finish care/cure and storage advice; keep hardware accessible for inspection, record faults and do not put an unsafe or uncertain chair into use. Formal approval remains separate from the student’s reflection.

Source: Project folio

Final quality checklist

Look for systematic documented observations and appropriate next actions, including unresolved faults. Teacher-directed checking controls practical acceptance. It is not a signed load certification, and touch-dry finish does not prove full cure.

Source: Project folio; Module 10, section 1; Module 10, section 3; Module 10, section 6